Our Brains in the AI Era

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DX at GitHub

I still consider myself a bit of an AI-skeptic

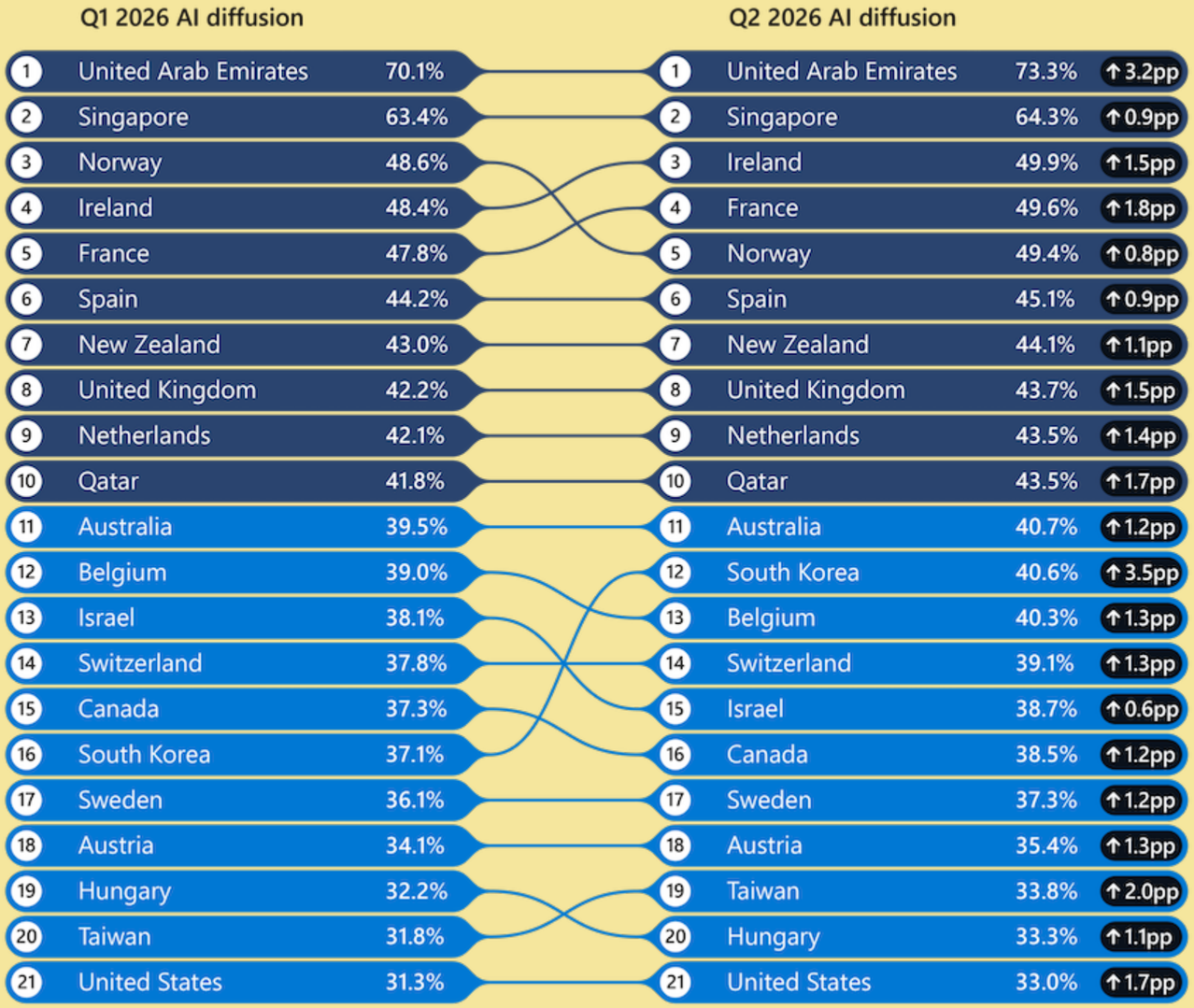
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cassidoo.co/ai

I work with AI users a *lot*

85% of developers use AI regularly[1]

[1] JetBrains' State of the Developer Ecosystem Report 2025



Microsoft Global AI Diffusion Report: aka.ms/AIDiffusionGlobal

How did we get here?

The waves of AI

The waves of AI

- "Pair programming", fancy autocompletion
- "Peer programming", agents
- "Orchestration", fleets of agents

At GitHub

At GitHub

Scaling Git infrastructure improvements

- More work in parallel
- Decoupling storage and compute
- Up to 35x the write throughput

At GitHub

Agent-generated PRs:
- January 2026: <10 million
- March 2026: 17.8 million
- August 2026: 71.1 million

At GitHub

Number of commits:

- 2025: 1 billion commits
- August 2026: 1.15 billion commits *per week*

What could possibly go wrong?

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- Debugging code you didn't write
- Defending decisions you didn't make
- *Too much* reduced **friction**

What could possibly go wrong?

In an MIT study, people wrote essays in groups, with and without LLMs.[3]

[3] gh.io/brain-decline

What could possibly go wrong?

- 83.3% of LLM users were unable to quote even one sentence
- 88.9% of "brain-only" + search users could quote accurately
- 0% of LLM users could produce a **correct** quote, while most "brain" users could

What could possibly go wrong?

- The story of a new learner
- OpenClaw writing a hit piece
- Job interview crutches
- Open source slop

“Once men turned their thinking over to machines in the hope that this would set them free. But that only permitted other men with machines to enslave them.”

— Frank Herbert, Dune

We have a duty in tech to be the stewards of the future.

Only ~16% percent of the world has ever used AI, ever.[4]

[4] global-ai-adoption.netlify.app

We need to build the future we want to be a part of.

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- Safety guardrails
- Sustainable tooling
- Accessible costs

We need to build the future we want to be a part of.

- Systems for communicating
- Forward-thinking hiring practices
- Mentorship pipelines

All of these concepts boil down to learning in public, more than ever.

Imagine your tools go down.

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- Could your knowledge get you through?
- Could your team, stakeholders, customers get you through?
- Could your records get you through?

Our brains are the only intelligence tools we'll have for our entire lives.

To **scale** our thinking, we have to **share** our thinking.

Thank you!

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Tell me your thoughts!

cass.run/cm